



Electrical behaviour of ionic liquid confined in nanochannels

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Abstract

The confinement of ionic liquid inside the nanochannels and its transportation through it under applied potential is being reported. The nanochannels in SiON/Si structure have been created by swift heavy ion bombardment of Au²⁶⁺ ions beam at a fluence of 10⁸ ions/cm² having 350 MeV energy. After chemical etching, these channels (~50 to 80 nm diameter) have been filled by ionic liquid (IL) to see their behaviour under applied electrical bias. Due to the high self-diffusion coefficient and low viscosity (compared to other ionic liquids), EMImTFSI has been chosen for the said purpose. The effect of temperature (35–50 °C) and concentration of ionic liquid (undiluted and 50%, 60% and 75% diluted) on IL-SiON/Si structures has been studied. Dissociation of the ionic liquid by the applied voltage and charge extraction/injection through ion channels results in diode-like current–voltage behaviour. Ion current rectification presented in this paper depends on ions concentration and the temperature of the ionic liquid. It also depends on the applied voltage at a given temperature. It has been found that the ion current rectification ratio decreases with dilution of the ionic liquid. For the system with 75% diluted IL, SiON/Si system has the same behaviour as the one without IL. Since the temperature affects the mobility of ions hence, it also affects the current–voltage behaviour of the system.

Keywords Ion channels · Ionic liquid · Ion current rectification

1 Introduction

Solid substrates (semiconductors, oxides, polymers) with nanochannels provide a wide range of applications in the field of electronics, biotechnology and chemical science. One of the effective ways to produce these nanochannels on a substrate is by swift heavy ion bombardment (Hoppe et al. 2008; Kaniukov et al. 2016; Nebogatikova et al. 2018; Rawat et al. 2010). Other techniques such as focused ion beam drilling, electron beam drilling and wet etching have also been used extensively in this regard (Tang et al. 2016). Fabrication of nanochannels via ion track etching techniques provide tuneable pore geometries such as conical, hour glass, funnel, cylindrical and dumbbell. Different geometries of the nanochannels can be used to study the transport phenomena of various types of molecules and nanoparticles in confined space (Ma et al. 2020). Different theoretical and computational methods to study the ion

transportation through various types of nanochannels have been discussed in detail by Modi et al. (2012). These types of artificial symmetrical and asymmetrical nanochannels created via ion track technology have various applications such as filters, sensors and bioinspired devices (Fink et al. 2005; Zhang et al. 2015). Due to the similarity of the nanochannels in PET (polyethylene terephthalate) membrane to the biological channels, they have great potential in the fields of biosensing, filtration and energy conversion. Similar technique has been used in the present study to fabricate asymmetrical nanochannels (conical) on SiON/Si substrate. The effect of confinement of ionic liquid on the ion current rectification and current–voltage behaviour has been studied. Chemical etching of the zones created by the ion bombardment results in nanochannels with high surface to volume ratio. Additional parameters such as channel density, their diameter and shape provide spatial distribution to the embedded material which influences the performance of the device made thereof (Fink et al. 2016; Negi and Chandra 2019; Rawat and Chandra 2011; Spohr et al. 1990). The confinement/transportation of various materials (conducting or magnetic) in/through the channels on solid substrates lead to various devices. One such device is voltage-gated

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engineered nanochannels. The study of ionic transport through nanochannels offers the possibility of boosting ionic transport to address/solve the issue of low ionic transport in electrochemical energy storage and conversion. Transport of molecules or ions through engineered nanochannels can lead to devices such as field effect transistor, ionic field effect transistor, ionic charge coupled device, ionic diodes, nanofluidic device, field effect enhanced energy conversion, biological and chemical analysis and energy conversion (Guan et al. 2014). Transport of ionic liquid ([BMIM⁺][SCN⁻]) and water mixture through conical nanochannels in PET exhibits nanofluidic memristor through which the ion transport/conductance depends on the history of ion flow (Sheng et al. 2017). The effect of channels' selectivity, applied electric field and temperature on the transportation and pumping of ionic liquid through nanochannels in different materials such as carbon nanotubes, graphene nanochannels, etc. have been studied methodically (Shin et al. 2015; Wang et al. 2019). The pumping of ionic solution is due to the movement of mobile ions gathered around the electrodes. The study of these parameters of the nanochannels in semiconductors are still to be explored. The channels created in the semiconducting material (SiON/SiO₂) can have tuneable properties giving the so called TEMPOS (Tuneable Electronic Material with Pores in Oxide on Silicon) structure-based devices. The relative low cost of fabrication along with compact size are the advantages of TEMPOS structures. The voltage-gated nanochannel structure studied in the present study is a three-terminal device that resembles a metal–oxide–semiconductor field effect transistor (MOSFET). These voltage-gated nanochannels integrate wet ionics with dry electronics. Different semiconducting, conducting and magnetic materials have been embedded in this structure to fabricate various nano electronic devices. In these devices, the ion channels in silicon oxide/oxy-nitride on silicon substrate have been used as charge extraction/injection channels (Fink et al. 2004, 2016; Liu et al. 2018; Rao et al. 2003; Saroch et al. 2008).

In the present study, ionic liquid (IL) has been incorporated in these nanochannels in SiON/Si structure to study the change in its charge transport behaviour due to confinement. Ionic liquids are solvent-free ionic fluids composed of ions only and have melting point below 100 °C. Ionic liquids can be designed appropriately choosing the cation–anion pair and hence are also known as designer solvents (Cerclier et al. 2014; Neouze et al. 2006; Zuo et al. 2019). Earlier, the confinement of ILs in polymer network, silica matrix, porous silica and carbon has thrown light on their glass transition temperature, conductivity and ion dynamics (Cerclier et al. 2014; Jitvisate and Seddon 2017; Neouze et al. 2006; Zuo et al. 2019). Confinement of ionic liquid preserves its properties and allows liquid-like behaviour at temperature below the crystallization temperature (Neouze et al. 2006). Dilution of these ionic liquids leads to crossover from well-ordered

ionic liquid to a well-ordered solvent (Jitvisate and Seddon 2017) and makes it easy to insert it inside channels. It has been found that when ionic liquid is confined in channels (charged/uncharged), its behaviour gets affected (Karnik et al. 2007; Jiang et al. 2016). The confinement of ionic liquid results in trapping of ions/molecules in the nanoporous material due to the interaction with the channel walls which changes the physiochemical behaviour of the ions. According to Karnik et al., spatial change in the surface charge along the length of channels on PET results in enhancement of the ionic concentration (Karnik et al. 2007). Different modification of the electrical double layer (EDL) structure near the channel tip by biasing voltage results in current rectification. Uniform and non-uniform surface charge pattern leads to rectification (Jiang et al. 2016). Limited literature is available reporting the confinement of ionic liquid inside semiconducting structure for device application (Pinilla et al. 2005; Wang et al. 2015). To understand the dynamical behaviour of confined ionic liquid in a semiconductor structure, its systematic characterization is necessary. Ionic liquid [1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl) imide, EMIm TFSI] has been inserted inside the nanochannels of SiON/Si structure and their current–voltage behaviour has been examined. The confinement of ionic liquid in the nanochannels within the semiconductor structure has shown current rectification due to its restricted transportation. These channels transport ions primarily in one direction while blocking them in the other direction. This type of ion current rectification has also been found in many engineered systems and has application in various types of biochemical sensing, ionic diodes and ionic circuits (Fink et al. 2007, 2011; Jitvisate and Seddon 2017; Hlushkou et al. 2012).

Deposition of ionic liquid inside the etched ion channels is a good choice as its resistance, thermal behaviour and current–voltage behaviour can be tailored. When ionic liquid comes in contact with an electrode, alternate layers of cations and anions (electrical double layer) form at the interface. The Helmholtz model explains the changes in potential in the ionic liquid sandwiched between an electrode and a semiconductor (Fujimoto and Awaga 2013). The inserted ionic liquid between the electrode and SiON/Si produces the electric double layer at the solid–liquid interface resulting in high-density charge accumulation in the ultrathin region. For designing a device using filled nanochannels, it is essential to understand the processes by which ions enter the channels. The present study has been based on the basic concept of ion transportation with applied voltage. When ionic liquid comes in contact with the electrode surface, there is substantial increase in the surface induced ion ordering region (Fujimoto and Awaga 2013). Closer to the electrode, a cation–anion alternate layered structure gets formed followed by a mixed cation–anion layered motif (Feng et al. 2010). During the insertion of the ionic liquid

in the nanochannels, a thin layer of ionic liquid also gets coated on the top of SiON/Si. This thin layer of ionic liquid on the SiON/Si structure plays an important role in the generation of charges for conduction. Most of the ionic liquids have electrochemical stability window up to ~ 5 V. When higher voltage is applied, the dissociation of ion pairs of the ionic liquid takes place (Vos et al. 2014). Experimentally, it has been found that ionic liquids have weak coordination between ions and their surrounding molecules (Tokuda et al. 2005). These ions get a path for charge extraction and injection through ion channels. As the size of ion channels is sufficiently larger than the ion pairs of the ionic liquid, there is a high possibility of confinement of a number of ion pairs of the ionic liquid inside the ion channels. In this structure, channels in silicon oxynitride behave as charge injection or extraction channels. Due to the controlled charge extraction or injection through these structures along with the interaction of the ionic liquid with the radiation damaged sites, current rectification is achieved. Previously reported studies of physicochemical properties of ionic liquid due to the temperature and concentration gradients have concluded that the density of ions decreases linearly with the temperature increase irrespective of the ionic liquid structure (Tokuda et al. 2005). The summation of the self-diffusion coefficient ($D_{\text{cation}} + D_{\text{anion}}$) at each temperature follows the order $\text{EMTFSI} > \text{EMIMBF}_4 > \text{BPTFSI} > \text{BPPF}_4$. The larger cationic size and larger cationic molecular weight may result in a higher viscosity of ionic liquids (Matthew et al. 2013; Noda et al. 2001). Also, the high concentration of IL leads to shorter electrical double layer formation at the electrode and pore walls which affects the rectifying properties of the device. Addition of acetonitrile or acetone increases the diffusion as well as ionic conductivity exponentially while it decreases the viscosity of the ionic liquid. To immobilize the charge carriers, a significant amount of molecular admixtures is required. Addition of acetone increases the ionic conductivity by almost 50 times (Chaban et al. 2012; Chaban et al. 2016). Keeping all this in view, the dilution and temperature dependent studies have been performed on the IL-SiON/Si systems.

The nanochannels present in the SiON have density of $\sim 10^8$ to 10^9 per cm^2 . Each nanochannel filled with ionic liquid behaves individually as a nanodevice and all these nanodevices function simultaneously parallel to each other. Ionic liquid inside the nanochannels makes them highly conducting and these parallel conducting channels act as charge extraction/injection channels. High applied voltages (upto 20 V) dissociate ion pairs and increase the number density of mobile ions which contribute to the conduction mechanism. Ion current rectification has been observed for the first time in the IL-SiON/Si based TEMPOS structure although ion current rectification of charged conical ion channels in polymeric membrane and asymmetrical geometrical nanopores

have been extensively studied since several decades (Davenport et al. 2009; Experton et al. 2017). The ion current rectifying phenomena occurs due to the asymmetric structure or asymmetric charge distribution within the nanochannels in SiON/Si. This study confirms that the EDLs in ionic liquid can greatly affect the transport phenomena and using asymmetrical ion channels one can achieve large rectification ratios.

Present study provides significant contribution towards the understanding of engineered nanochannels based structures and related properties such as ion current rectification, EDL formation and effect of potential, concentration and temperature on their characteristics. These structures have future application as diodes in electronic circuits and energy conversion applications which can be tuned by varying concentration and temperature.

2 Material synthesis

Silicon oxynitride (amorphous) layers of ~ 500 nm thickness were deposited onto the top of the p-doped silicon wafer of ~ 375 μm and of resistivity of $35 \Omega \text{ cm}$ cut in [100] direction (Santiago 2019). The silicon wafer acts as carrier material. After cleaning, it has been oxidised in dry atmosphere (1200°C). Subsequently, the top layer has been protected by spin coated back photo-resist film and the rear side has been washed away with ammonium fluoride etchant solution. Finally, Al layer has been evaporated on the rear side to serve as the back contact. The SiON/Si substrate has been irradiated with 350 MeV Au^{26+} ion beam of fluence $10^8/\text{cm}^2$ at the swift heavy ion accelerator (ISL) at the Helmholtz Centre for Materials and Energy (erstwhile Hahn Meitner Institute), Berlin, Germany. This creates latent ion channels of nano-dimension on the SiON layer. The irradiated zones have been further etched chemically using 4% HF solution for 7 min at room temperature after which the substrate has been thoroughly rinsed with distilled water to stop the etching process. The etching process creates well defined conical channels of diameter ~ 50 – 80 nm that are parallel to each other and suitable for the insertion of conducting ionic liquids. In the present study, two aluminium electrodes of diameter ~ 4 mm have been deposited on the SiON top surface with separation of 7 mm with the help of vacuum coating unit. In the present work, AFM (Veeco 2002, non-contact mode, Si_3N_4 tip) has been used for imaging.

The energy loss during ion-beam irradiation process has been evaluated using SRIM (Stopping and Range of Ions in Matter) program (Ziegler and Biersack 1985). For the ion energy of 350 MeV, electronic stopping loss dominates ($9.237\text{E}+01 \text{ MeV}/(\text{mg}/\text{cm}^2)$) over the nuclear stopping ($3.959\text{E}-01 \text{ MeV}/(\text{mg}/\text{cm}^2)$) and the projected range shows that the target Au ions penetrate $\sim 39.74 \mu\text{m}$ in the SiON

layer on Si substrate. As the Au ions create nanochannels of depth $\sim 40\ \mu\text{m}$ and the diameter of the channels is ~ 50 to $80\ \text{nm}$ (by AFM imaging), the aspect ratio for nanochannels obtained is ~ 800 to 500 .

Ionic liquid 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide (EMImTFSI) of MW 391.31 has been purchased from Sigma–Aldrich. Dilution of IL (50%, 66% and 75%) has been done using acetone for better insertion inside ion channels. Ionic liquid of different concentrations has been inserted using spin coating unit (750 rpm, 500 s). The pH of EMImTFSI is ~ 7 and the dilution has been done by acetone which does not change the pH. A thin layer of ionic liquid has also been coated on the top of the substrate during the process of insertion. To study the effect of temperature (since the rise in temperature changes the viscosity of inserted ionic liquid inside ion channels) and dilution on the confined ionic liquid, current–voltage measurement of undiluted and diluted ionic liquid filled structures have been done.

2.1 Measuring set up

The fabricated IL–SiON/Si structures have one back and two surface contacts. In the present work, the surface contacts A and B have been used to apply the potential and the contact C is kept at ground potential, Fig. 1a. For the temperature related current–voltage measurements, the structure has been kept on a hot plate and heated to desired temperature. For concentration related measurements, the IL diluted by acetone has been inserted inside the ion channels and the fabricated structure has been characterized using current–voltage measurements.

3 Results

3.1 Current–voltage behaviour

The mechanism related to the transport of charge carriers leading to the conduction process has been studied using current–voltage (I – V) measurements. Current–voltage measurement has been performed using the top electrodes denoted by A and B in Fig. 1a. AFM 3D image of the etched nanochannels on SiON/Si for the area $500 \times 500\ \text{nm}^2$ has been shown in Fig. 1a. It is seen in Fig. 1a that the swift ions create nanochannels in the substrate having a diameter of $\sim 50\ \text{nm}$ which increases to ~ 70 to $100\ \text{nm}$ for overlapping channels. The image is of the top surface of the substrate and the density of channels in SiON/Si is $\sim 3 \times 10^8/\text{cm}^2$. The peak and roughness analysis has been done with Gwyddion software. Table 1 shows the line analysis of the peaks according to their position. The peak depth is $\sim 7\ \text{nm}$ and the average roughness (R_a) is $\sim 2.77\ \text{nm}$ and root mean

square roughness (R_q) is $\sim 3.84\ \text{nm}$. These channels are distributed over the entire surface of SiON/Si. The current has been measured using $6\frac{1}{2}$ digital multimeter with applied voltage ($-20\ \text{V}$ to $+20\ \text{V}$) at 10-s intervals. EMImTFSI ionic liquid has electrochemical stability window of $\sim 5\ \text{V}$. When in contact with an aluminium electrode, the stability window is reduced due to the dissociation of aluminium (into Al^{3+} ions). The applied voltage ranges in the present study ($-20\ \text{V}$ to $+20\ \text{V}$) causes electrochemical reactions to occur at the electrodes (despite the fact that the TFSI anions can passivate the Al^{3+} ions), thereby leading to ionic liquid dissociation at the negative electrode and aluminium dissolution at the positive electrode (Fekete et al. 2016; Kühnel et al. 2012). Dissolute aluminium reacts with the TFSI ions present in the ionic liquid and the cations of EMIm move towards the negative electrode. The flow of the ions depends on the applied voltage. High applied potential increases the dissociation of ion pairs generating more charge carriers (“free” ions). The dissociated ions take part in the conduction mechanism and the increased number density of mobile ions results in conductivity enhancement. The higher applied voltages result in higher polarization charge densities of the ions of the ionic liquid in the nanochannels and thus, an increased conductance through the nanochannels which results in higher ion current rectification. Figure 1b shows the current–voltage behaviour of the SiON/Si (empty channels) and IL–SiON/Si samples. As seen from Fig. 1b, the current–voltage characteristics of SiON/Si substrate with empty ion channels is ohmic in nature while ion channels filled with ionic liquid shows non-symmetric and nonlinear behaviour. It also shows reduction in the current value as compared to the SiON/Si substrate. This lower value of current is likely due to the degradation of IL at the aluminium electrode (at negative potential) forming an insulating layer (Cabana et al. 2018). In the forward direction, a linear variation in the current with voltage has been obtained while in the reverse direction, almost negligible current flows. Although ion channels act as charge extraction/injection channels, the confinement of undiluted (highly viscous) ionic liquid inside them exhibits linear resistive behaviour as the high viscosity of the IL restricts ion motion in the confined channels.

As the current–voltage (I – V) behaviour cannot distinguish the refinement(s) in the ion current behaviour if at all and represents the average charge transport, current–time measurement (I – t) has also been done. The variation of current with time is shown in Fig. 1c. These data have been obtained for applied voltage ($-2\ \text{V}$, $-1\ \text{V}$, 0 , $1\ \text{V}$ and $2\ \text{V}$) in steps of $0.5\ \text{s}$ for a total time of $500\ \text{s}$. I – t behaviour indicates that the current is constant with time for all the applied voltages. The stable current shown in the current–time plots in Fig. 1c support the fact that there is redox chemistry occurring at the electrodes. As ILs are by themselves electrically insulating,

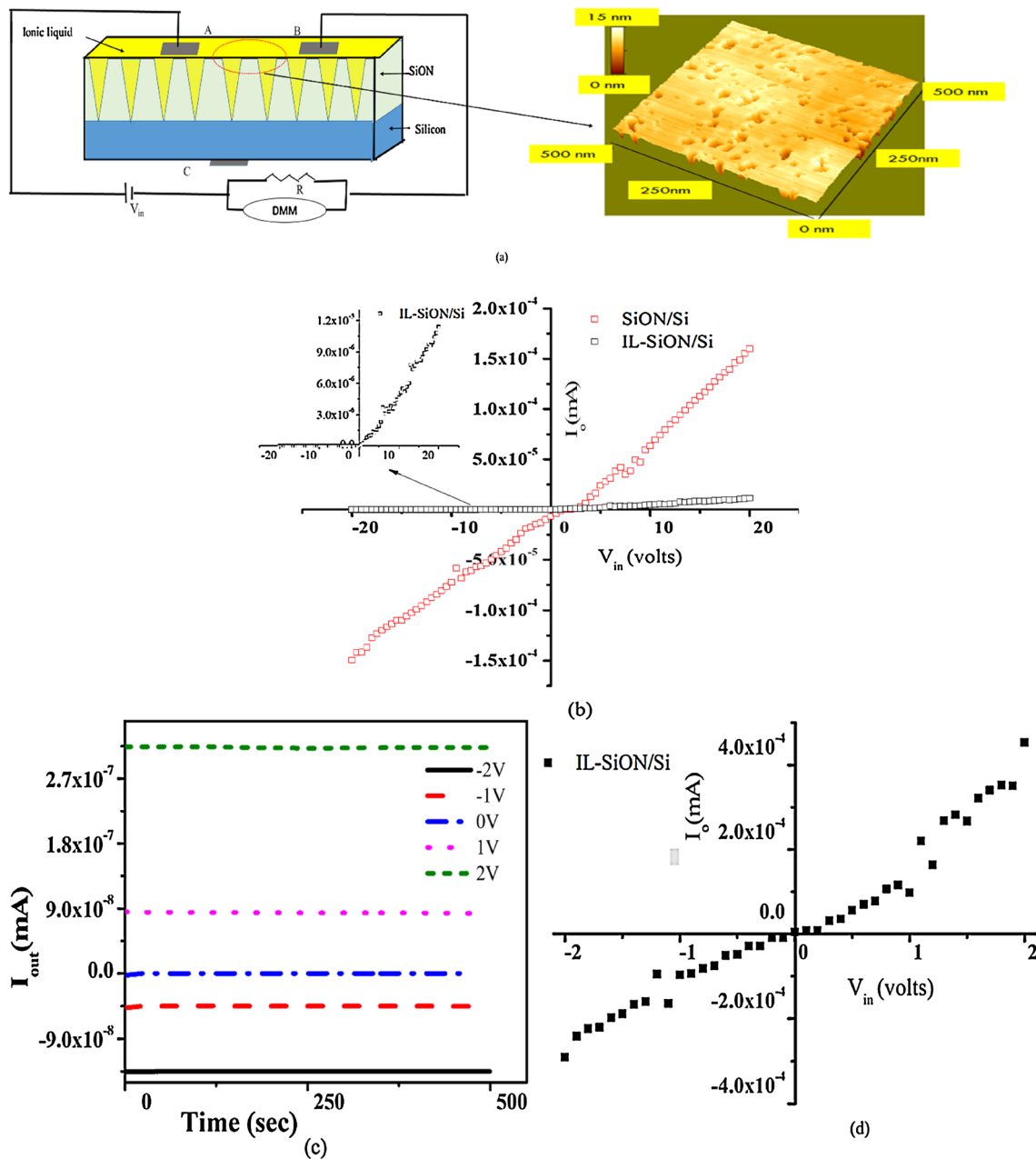


Fig. 1 **a** Experimental circuit, **b** current–voltage behaviour of empty and IL filled substrate, **c** current–time series with applied voltage -2 V to $+2$ V and **d** current–voltage behaviour of IL-SiON/Si at applied voltage (-2 V to 2 V) at 10-s intervals

they are unable to contribute to the current in the absence of redox reactions occurring at the electrodes. To further verify this phenomena, current–voltage of IL-SiON/Si structure at lower applied voltages has also been measured and shown in Fig. 1d. As it can be seen from Fig. 1d, the current–voltage behaviour is ohmic similar to I – V behaviour of SiON/Si (empty channels). This indicates that at higher applied voltage, electrochemical reactions occur at the electrodes due to the dissociation of the ionic liquid which results in flow of the ions through the channels.

The charge carrier behaviour depends on the geometrical configurations of the channels (shape, length and diameter) of the TEMPOS structure. Channels created in the SiON/Si are conical with outer diameter ~ 50 to 80 nm (inset Fig. 1a). This enables the ionic liquid to easily enter and get trapped inside them. As a layer of ionic liquid also gets coated on the upper surface of SiON, the applied potential is also distributed through this layer. Ions of the ionic liquid collect at the counter electrode as shown schematically in Fig. 2. These interfacial double layers are most apparent in systems with

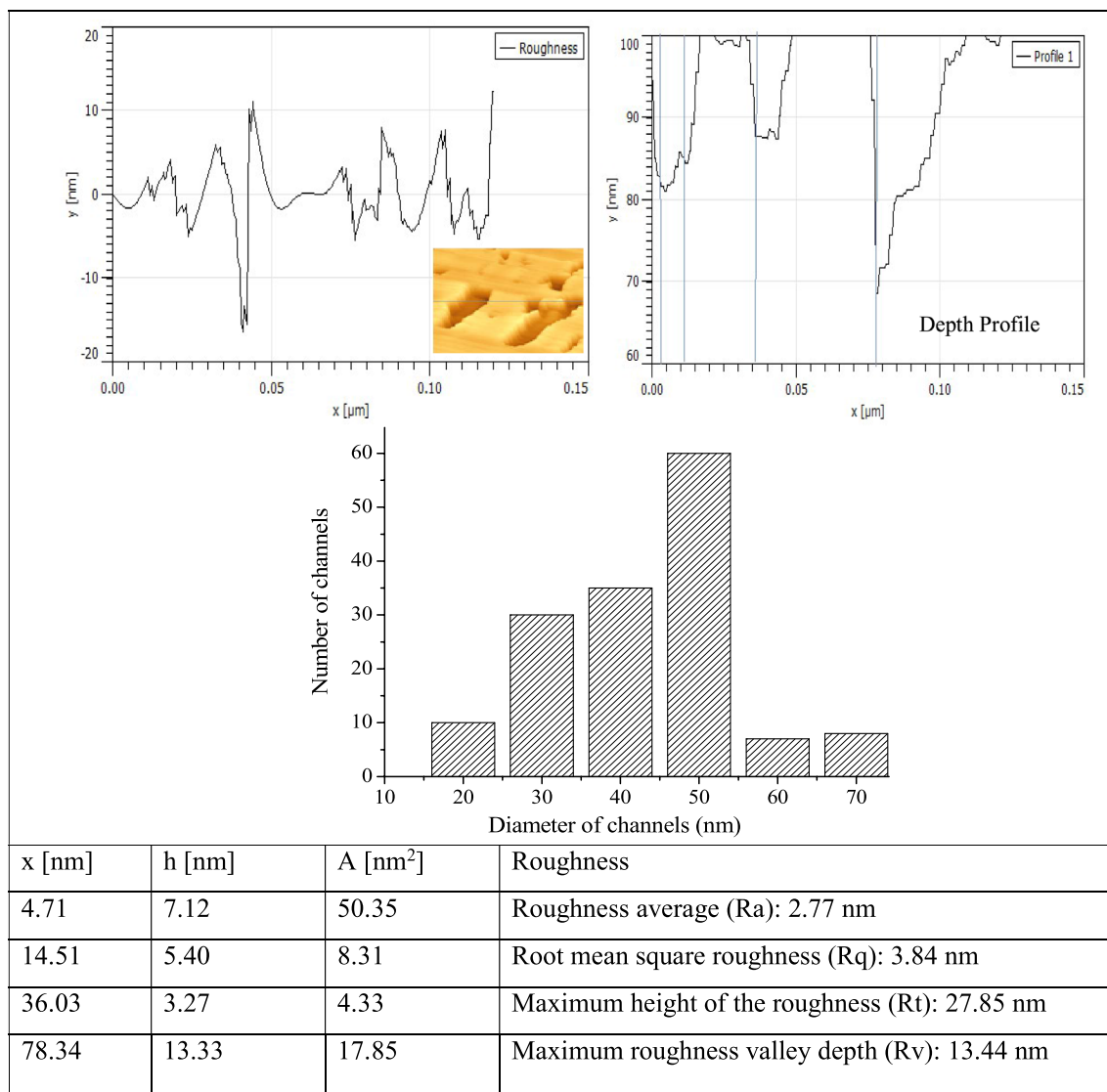
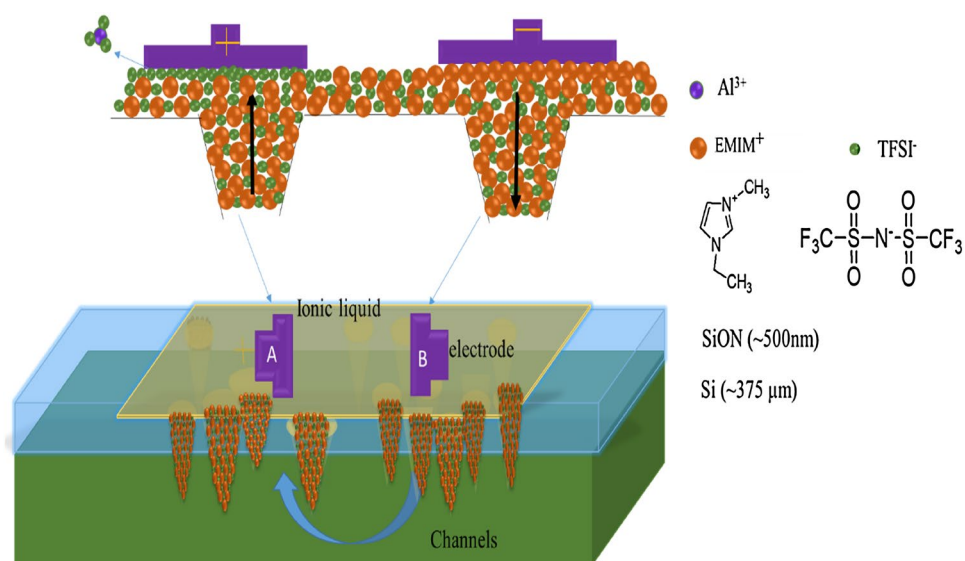
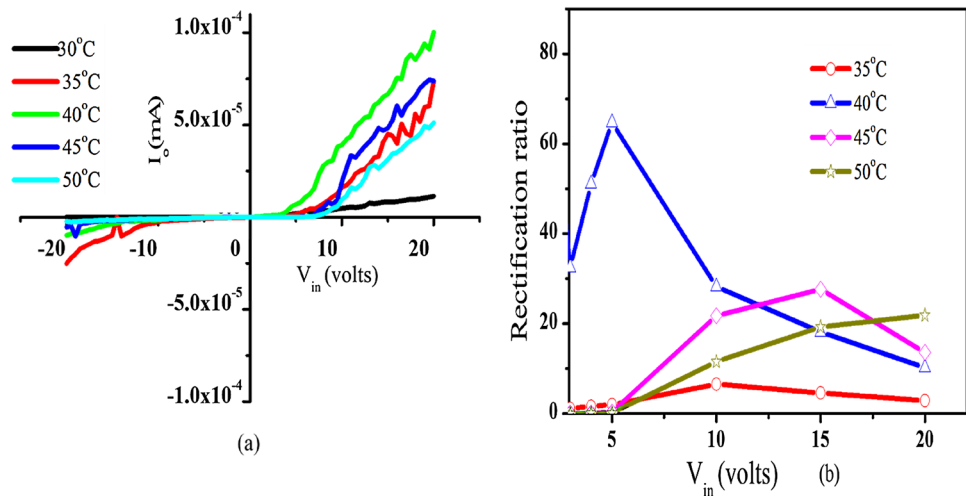
Table 1 Line analysis, roughness and distribution of the nanochannels' size (in $500 \times 500 \text{ nm}^2$)**Fig. 2** Schematic representation of charge injection and transportation in IL/SiON/Si system

Fig. 3 **a** Current–voltage measurements of IL filled TEMPOS structure at different temperatures and **b** ionic current rectification ratio vs voltage for different applied voltage



large surface to volume ratio as is the case in the present IL-TEMPOS structure. Under the influence of applied voltage, charges are injected from the surface IL layer into the channels and extracted from channels at the silicon substrate [38]. The charges injected at the contact point on the surface layer of the IL will be spread around the electrode (radius of influence ' r '). These spread out charges will enter the filled etched channels below them (or in close proximity to them) and subsequently be driven through the channel towards the conducting Si substrate. It is well known in ILs that dissociated ions form ion pairs which affect the conductivity of the ILs. As the ion pair diameter of EMImTFSI is ~ 0.75 nm (Jiang et al. 2016), there is a high probability of the presence of a large numbers of ion pairs inside the channels. In the present study, as the channels have been created by swift heavy ions and are randomly distributed, the probability of charge distribution will always be high for small values of ' r '.

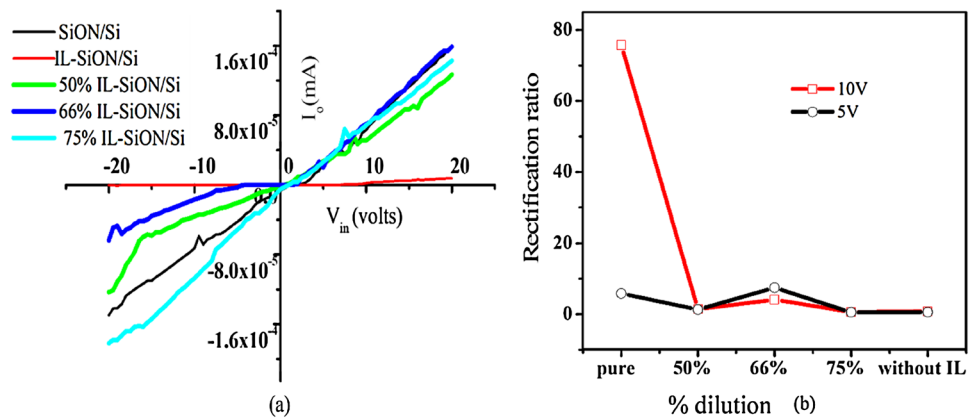
3.2 Effect of temperature

The temperature affects the physical properties as well as the ion transport properties of the ionic liquid. It is known that the ionic liquids are molten around room temperature limiting the operating temperature which should be kept above their melting temperature to avoid their solidification. Ionic liquid (EMImTFSI) used in the present study has melting temperature of ~ -16 °C. In this section, the effect of temperature on the IL filled TEMPOS structure has been discussed. IL-SiON/Si structures have been heated from 30 to 50 °C and respective current–voltage measurements have been done. Due to the effect of temperature, the viscosity of the IL changes which affects the flow of ions on the top layer as well as inside the channels. Figure 3a shows the current–voltage behaviour of heat treated IL-SiON/Si structures. As seen in Fig. 3a, the temperature rises from

30 to 50 °C results in change in the I – V characteristics of IL-SiON/Si structure from ohmic (at room temperature) to rectifying/diode-like (at higher temperatures). The forward/reverse currents increase initially with temperature up to 40 °C and then decreases. It should be noted that the cut-in voltage shifts to higher voltage for this very reason. As the temperature affects the viscosity and mobility of the charge carriers of the ionic liquid, the rate of electrochemical reactions is also affected which is evidenced by the I – V behaviour. Initially, the current increases as the viscosity of the ionic liquid decreases with temperature (till 40 °C). Thereafter, as the temperature increases further, the current decreases due to the collection of ions in the channels which arrive there in large numbers due to gain in mobility as a consequence of thermal energy. This hinders the flow of ions and hence current through the channels.

As the current–voltage behaviour gets affected by temperature, to quantitatively analyse the current rectification, the rectification ratio (RR) has been determined with applied voltage for different temperatures. The rectification ratio is defined as the ratio of the currents at particular value for (both positive and negative) applied voltage, i.e., $RR = \left| \frac{I(V)}{I(-V)} \right|$. Thus, the RR is dependent on the applied voltage. Here, the RR values corresponding to the Fig. 3a have been calculated. Figure 3b shows the variation in the rectification ratio with the applied voltage at different temperatures. As is seen from figure, the rectification depends on the applied voltage at a given temperature. The ion current rectification results due to the asymmetric shape of the channels which gives rise to the internal asymmetric electrostatic fields (Fink et al. 2011). The rectification which can be observed as asymmetric current–voltage behaviour, current higher for one polarity compared to the other, indicates preferential direction for ion flow. Ion current rectification is also a consequence of asymmetric distribution of surface charges/

Fig. 4 **a** Current–voltage behaviour with concentration variation and **b** rectification ratio vs % dilution



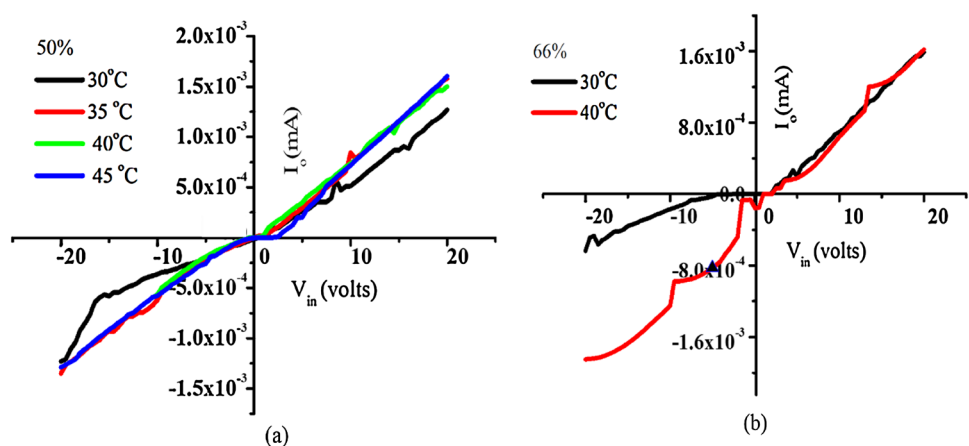
uneven ionic strength of channels. It has been reported that there is high positive charge ($\sim 10^{12} \text{ cm}^{-2}$) on SiON and SiON/Si interface (Zhuo et al. 2013). This surface charge results in the accumulation of anions of the ionic liquid on the walls of the channels and EDL formation. Due to uniform distribution of surface charges, EDL formation is uniform. The EDL of the ionic liquid consist of two layers; the inner layer of closed packed counter ions and outer diffusive alternate layers of anions and cations. The EDL generated near the channels has an influence in the ion transport through the channels. The ion current rectification also strongly depends on the channel opening and the ionic concentration of the solution (Mirkin and Amemiya 2015). As the ion flow depends upon the applied voltage, concentration and temperature, the rectification ratio would also be affected by these parameters.

3.3 Effect of concentration

One more factor that could influence the properties of IL-SiON/Si structure is concentration of the ionic liquid. The high IL concentration (with large no. of mobile ions) leads to high viscosity of ionic liquid which would lessen the electrical conductivity. Dilution decreases the concentration while

increasing the mobility of the charge carrier. For dilution of EMImTFSI, acetonitrile, acetone and water have been used by different workers (Noda et al. 2001; Mathew et al. 2013; Chaban et al. 2016). In the present study, IL has been diluted using acetone. The low viscosity of acetone facilitates drastic viscosity decrease of EMImTFSI. Due to the carbonyl group present in acetone, it is easily miscible in ionic liquid. Acetone also exhibits low viscosity (0.307cP) and high diffusion. Diluted ionic liquids (50%, 66% and 75%) have been inserted in channels and have been compared with undiluted IL-SiON/Si structure at RT. Figure 4a shows the current–voltage behaviour of diluted IL-SiON/Si structure. Dilution of ionic liquid would not only enhance the electrical conductivity but also affect the interaction of ions with the channel wall. Concentration of ionic liquid would affect the Debye length as higher concentration leads to shorter Debye length. Hence, the dilution will increase the Debye length and thereby, the current value. The interaction of ions of the ionic liquid with aluminium electrode and the ion flow are also influenced by the concentration of the ionic liquid. It can be seen from Fig. 4a, b that the rectification is also affected due to the dilution and the current value changes as more dilution leads to flow of large number of ions. Figure 4b shows the effect of dilution on the rectification

Fig. 5 Effect of temperature on IL-SiON/Si system's current–voltage behaviour for **a** 50% dilution and **b** 66% dilution



ratio (RR) of the IL-SiON/Si structure for different applied voltages (5 V and 10 V). For undiluted IL based systems, higher applied voltage results in higher polarization charge densities in the nanochannels and thus, an increased conductance through the nanochannels. The high ion current rectification results from more polarization in combination with an asymmetric surface charge distribution. Increasing the voltage first causes diffusion limited current regime and then there is an enhanced flow of current which results in the high rectification ratio (Kim et al. 2007; Yossifon et al. 2009). The rectification ratio decreases with dilution and for 75% diluted system it behaves the same as SiON/Si system (without IL) as seen in Fig. 4b. The current–voltage behaviour and the current value of SiON/Si and 75% IL-SiON/Si is almost the same. This behaviour has been attributed to the very low concentration of ions in the upper layer and inside the channels. The 66% diluted IL-SiON/Si system has been found to have high rectification as well as high current value compared to other diluted systems.

As discussed above, the temperature also affects the system like dilution. Heat treatment of diluted 50% IL-SiON/Si and 66% IL-SiON/Si systems leads to a change in their I – V behaviour. Figure 5a, b show the effect of temperature on the diluted IL-SiON/Si systems. As seen in the figures, the heating of 50% diluted system at 30–45 °C results in change of the I – V and also affects the rectification behaviour. For 66% diluted system, ohmic behaviour is obtained at 40 °C.

4 Conclusion

Above experimental findings confirm the rectifying behaviour of ionic liquid confined inside channels in SiON/Si system. The tuning of rectification and current–voltage behaviour can be achieved by concentration and temperature variation. The ionic liquid layer present between the electrode and the ion channels' opening is responsible for this behaviour. As the thin IL layer gets dissociated into cations and anions which start flowing through the ion channels, it results in rectifying I – V behaviour. The maximum rectification has been achieved for undiluted IL-SiON/Si structure at room temperature. The merit of the system is that a small amount of IL is used which also reduces the required processing time. The SiON/Si structures with nanochannels have applications as nanofluidic ionic diodes, ionic switches and sensors. Although a prototype can be obtained in the laboratory, the key challenges in mass scale production of such devices are (1) precise and controllable etching of nanochannels, (2) difficulty in obtaining different surface charges in a nanochannel and (3) limited process integration ability.

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

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